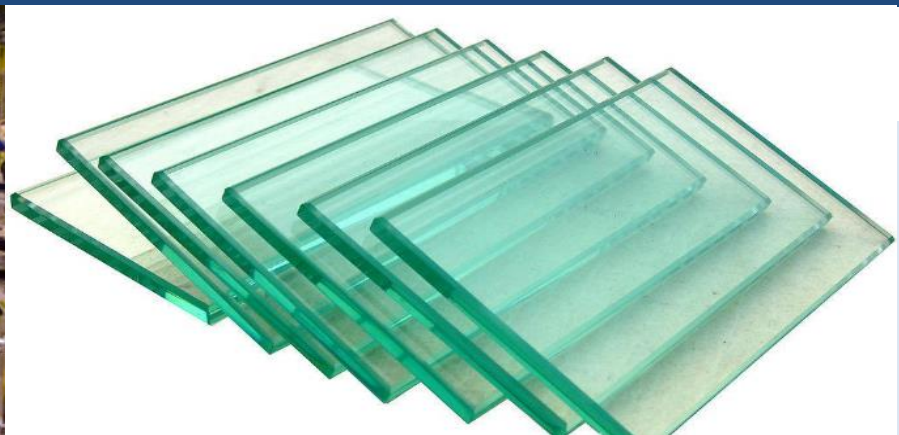


A Pre-feasibility Study for the Establishment of a Glass Factory in Ma'an



2017



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1. Executive summary

In Jordan, there are many factories that produce glass and mirrors, that imported glass from outside of Jordan, due to the lack of this product in the country, and that's why the idea to have glass factory in Jordan came to life.

This project aims to supply the Jordan and global market with glass because of the strong demand for glass. In addition, it aims to export to the other market.

Additionally, the idea is to manufacture raw glass and sell it to companies, glass factories and to the Jordan market. Therefore, the location of this factory has been selected to be in Ma'an because there are many raw materials, particularly glass sand in this Governorate, which is the main component of glass.

Based on the proposed market share and prices that have been adopted by the proposed project, the expected revenue amounted to 882,000 Jordanian dinars during the first year of the project and gradually rise to reach 1.47 million Jordanian dinars during the tenth year. Furthermore, the period to recover the amount of capital is equal to six years with an internal return rate equal to 20.06%.

The economic indicators of the project are shown in the table below:

Table 1: Economic indicators

Indicator	
Total Investment Requirement (JD)	2,170,965
Internal Rate of Return (IRR)	%20.06
Pay-Back Period (year)	6
Net present value (JD)	576,912
Job Opportunities provided by the project	30
Annual Revenue Projections (10 years)	12,936,000
Total Operating Expenses (10 years)	6,360,245
Net Profit (10 years)	1,954,920

2. Key Highlight of Jordan

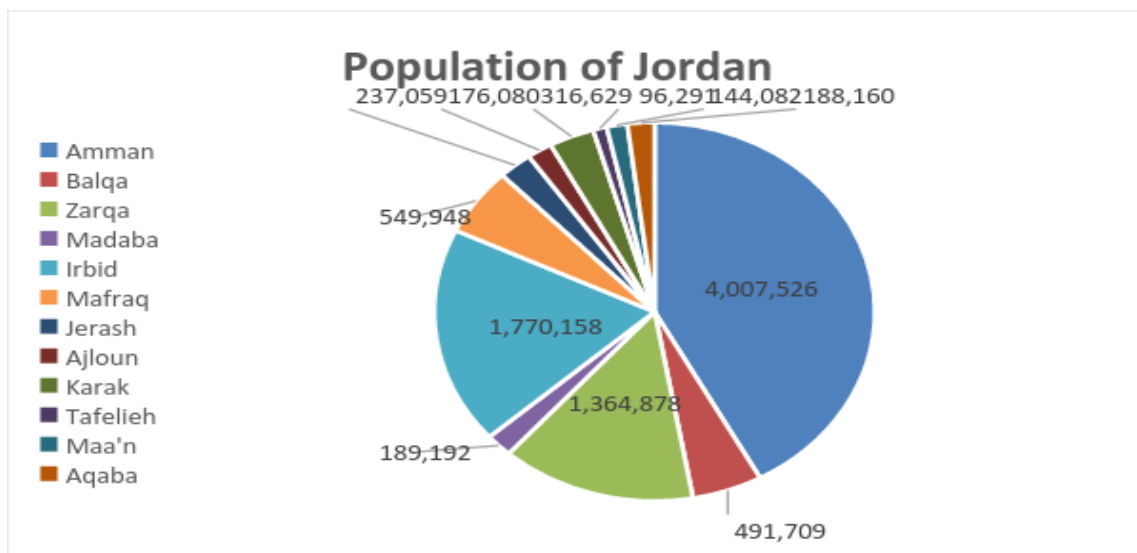
Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates. Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafrq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table 2: Population Distribution according to Age Group

Age Group	% Population	Males	Females
Years 14-0	35.04%	19.2%	18.2%
Years 64-15	61.02%	30.7%	28.6%
More than 65 Years	3.94%	1.6%	1.6%

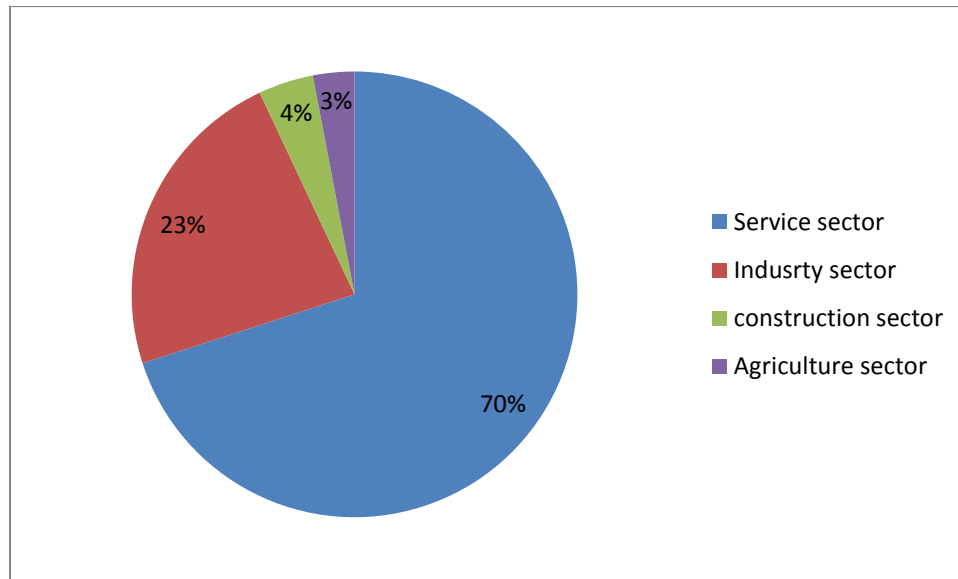
Department of Statistics (DOS), 2015

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure2 Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and comprising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

Furthermore, in its endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

3. Market analysis

Project Description

The project aims to establish a glass factory in Ma'an, with the location selected due to the large availability of raw materials, which is the main component of the glass panels' product. The most important material of these raw materials is glass sand "silica", which is available particularly in Ra'as Al-Naqab and in Ma'an generally.

The factory will cut, rinse and manufacture glass with different sizes, which are used in doors and windows of shops and buildings and companies, and are used in cladding of the facades of buildings and shops because of the beauty of its form, ease of cleaning, endurance and isolation of heat

Glass has many characteristics represented in transparency, hardness and resistance to chemicals.

Project Objectives

Project objectives are as follows:

- Exploit the natural resources available in Jordan for the glass plates production.
- To achieve profitable returns for project owners.
- Production of high-quality glass panels meets the demands of the Jordan market and external market.
- Provide job opportunities and reduce unemployment rates.

Proposed Product

Glass is a colorless substance mainly manufactured from melted silica at high temperatures with boric acid or phosphate. Basic components of glass are from silica sand, flint and quartz. Silica is dissolved in very high temperatures to produce molten silica glass.

Different types of silica glass are produced as a result of silica interaction with other raw materials in different proportions. In addition, there are alkaline compounds such as sodium carbonate and potassium carbonate that reduce the degree of melting temperature and the viscosity of silica.

Glass melts usually at a high temperature and does not expand or shrink significantly with temperature change, and then so it will be appropriate for glass panels production (final product).

Target Segments

Target groups are:

- Glass shops
- Factories and shops selling aluminum doors and windows.
- Construction companies and offices, where glass panels are used to cover the facades of buildings.
- Factories of glass vehicle Production

Location Analysis

Al- Rawda Industrial City is the main area in the Ma'an development area, which covers an area of 2.5 square kilometers, 750 thousand square meters of the city were developed, which are prepared with a full infrastructure necessary to receive industrial investments.

Al- Rawda Industrial City is located 10 km south-east Maan, which is well suited to provide the requirements of establishment and running the light, medium and heavy industries. Therefore it has become the headquarters for many factories in the field of construction, electrical and chemical.

Market Size Analysis

Table 3: Volume of exports and imports of glass for the year 2015

Commodity_Description	Imports: value of the Jordanian dinar	Import: Quantity	Domestic Export: Value F.O.B. in J.D.	Domestic Export: Quantity	Re-Export: Value F.O.B. in J.D.	Re-Export: Quantity	Total Export: Value in J.D.	Balance (Total Export - Import): Value in J.D.
Non-wired glass, having an absorbent, reflecting or non-reflecting layer	6,351,025	26,342,071	649,935	1,156,170	126,021	317,965	775,956	-5,575,069
Coloured throughout the mass (body tinted), opacified, flashed or merely surface ground	5,329,320	22,965,444	16,758	25,627	0	0	16,758	-5,312,562
float glass and surface ground or polished glass, in sheets, non-wired, (other than coloured throughout the mass (body tinted), opacified, flashed or merely surface ground)	3,638,600	15,313,067	203,010	375,325	139,996	225,136	343,006	-3,295,594

Commodity_Description	Imports: value of the Jordanian dinar	Import: Quantity	Domestic Export: Value F.O.B. in J.D.	Domestic Export: Quantity	Re-Export: Value F.O.B. in J.D.	Re-Export: Quantity	Total Export: Value in J.D.	Balance (Total Export - Import): Value in J.D.
float glass and surface ground or polished glass, in sheets, wired glass, whether or not having an absorbent or reflecting layer	946,962	3,852,273	0	0	0	0	0	-946,962

The previous table shows that the volume of imports of glass exceeds by a large margin the volume of exports. This indicates that the market needs a glass factory able to cover the huge demand, where it enters the production of many local products.

Moreover, the Jordan market size of glass for 2015 is about 66,372.6 tons, which is equivalent to 3,318,630 square meters of glass. It is noted that the market size and the consumption rate of glass are growing continuously, as there is continuous rise in construction and increase in demand for glass panels by the construction and reconstruction companies.

Project Market size depends on building and construction activity in Jordan, where modern construction is moving towards the use of glass instead of stone, due to the low cost, beauty of the view and ease of cleaning.

The following table shows the number of construction licenses and construction areas in Jordan during the period (2012-2016):

Table4 : Construction size in Jordan

Period	2012	2013	2014	2015	2016
For residential					
Number of Licenses	29,940	33,906	36,642	32,882	36,367
Amman	7,956	8,096	8,904	7,596	7,172
Irbid	8,429	10,063	11,118	10,496	11,293
Zarqaa	2,804	3,223	3,103	2,765	3,292
Others	10,751	12,524	13,517	12,025	14,610
Area (thousands)	10,691.2	11,817.6	12,578.4	10,879.0	10,799.7
Amman	6,013.2	6,075.6	6,084.1	4,967.3	4,019.7
Irbid	1,806.0	2,243.1	2,611.9	2,378.1	2,599.3
Zarqaa	756.3	942.9	983.4	888.8	1,076.3
Others	2,115.7	2,556.0	2,899.0	2,644.8	3,104.3
other purposes					
Number of Licenses	2,526	2,695	2,936	2,893	3,043

Glass factory - Ma'an

Period	2012	2013	2014	2015	2016
Amman	1,188	1,176	1,334	1,322	1,373
Irbid	431	568	672	657	547
Zarqaa	338	423	471	477	600
Others	569	528	459	437	523
Area (thousands)	2,215.5	2,167.5	2,413.9	2,243.8	2,510.5
Amman	1,604.0	1,338.3	1,616.8	1,457.0	1,618.5
Irbid	131.1	254.8	233.6	268.9	188.6
Zarqaa	146.6	175.6	265.2	217.3	294.4
others	333.8	398.8	298.3	300.6	409.0

Source: Central Bank / Monthly Statistics / 2015

Glass is used extensively in different kinds of vehicles. The market size depends on the number of vehicles in Jordan, where the increase in the number of vehicles leads to an increase in demand for glass.

Table 5: Number of licensed vehicles and Change rate

Year	Number of vehicles	Percentage of increase%
2004	614,614	%8.5
2005	679,731	%10.6
2006	755,477	%11.1
2007	841,933	%11.4
2008	905,592	%7.6
2009	994,753	%9.9
2010	1,075,453	%8.1
2011	1,147,258	%6.7
2012	1,213,882	%5.8
2013	1,263,754	%4.1
2014	1,329,888	%5.2
2015	1,411,666	%6.1

Based on the Department of Statistics, the number of road accidents amounted to almost 115,000 accidents divided into collision and deterioration .

The expected number of vehicles that change glass is 140.000 vehicles annually based on the assumption that 7% of the vehicles changing glass as a result of annual accidents.

Competitors Analysis

This project is the first project of its kind on the level of Jordan and there is no existing plant currently that manufactures glass. Therefore all flat glass panels companies and factories importing the raw materials from outside Jordan, the most important of these factories in Jordan are as follows:

- 1- Al-Faris for glass industries.
- 2- Ezzat Mansour for glass.
- 3- Al-Awla for glass industries.
- 4- Al-Esraa' for glass industries.
- 5- Many small factories and places of sale and formation of the glass.

Price Analysis and Pricing Policy

Glass panels imported from outside Jordan, and Saudi Arabia is one of these countries that Jordan exports from. The following table shows the prices of proposed glass in Jordan market per square meter:

Table 6: Price Analysis

Type of glass	Price (Jordanian Dinar)
Glass 6 mm	2.50
Glass 8 mm	3.75
Glass 10 mm	4.75

Expected Occupancy Rates

The plant productive capacity is 8,000 tons per year, equivalent to 400,000 square meters, where one ton equals 50 square meters. Furthermore, the market share expected to be 12% of the local market size with the idea toward export to outside of Jordan in the future, where the global glass market is in high demand and is growing from year to year.

Glass will be produced in the following sizes:

Table7 : Percentage of each product from the total production

Type of glass (mm)	Percentage from the total production
6 mm	30%
8 mm	40%
10 mm	30%

Projected Revenues Analysis

The following table shows the expected revenues for 10 years from the start date of the project:

Table 8: Project revenues from 2018-2027

Prices assumptions	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Production percentage from total production capacity	60%	75.000%	85.000%	85.000%	85.000%	95.000%	95.000%	100.000%	100.000%	100.000%
Revenues of glass 6 mm	180000	225000	255000	255000	255000	285000	285000	300000	300000	300000
Revenues of glass 8 mm	360000	450000	510000	510000	510000	570000	570000	600000	600000	600000
Revenues of glass 10 mm	342000	427500	484500	484500	484500	541500	541500	570000	570000	570000
Total Revenues	882000	1102500	1249500	1249500	1249500	1396500	1396500	1470000	1470000	1470000

The projected annual revenues for the project were calculated based on the project productive capacity analysis as the following table shows:

Table9 : Production capacity Analysis

Production capacity Analysis	
Annual production capacity	400,000
Price per square meter of glass 6 mm (JD)	2.500
Price per square meter of glass 8 mm (JD)	3.750
Price per square meter of glass 10 mm (JD)	4.750

Glass factory - Ma'an

4. Technical analysis

Required Human Resources

Table10 : required Manpower

Job title	Number
Indirect employees	
General Manager	1
Manager Assistanat	1
Engineers	1
Human Resources Manager	1
Labortary worker	5
Sales representatives	5
Financial Manager	1
Accountant	2
Human Resources Employees	2
Total Number of indirect employees	18
Direct employees	
Quality Control Managers	2
Maintenance technician	3
Warehouse manager	1
Electrical technician	2
Mechanical technician	2
Quality Control Engineers	2
Total Number of direct employees	12
Total Employees	30

Job Description:

- **General Manager:** Will propose the public policy of the Plant and the preparation of plans and follow up on their implementation, technical and administrative supervision of staff, and development of procedures and ensuring occupational health and safety requirements.
- **Financial Manager:** Will be responsible for all financial transactions in the plant.
- **Accountant:** Responsible for tracking of financial issues auditing.
- **Marketing officer:** Responsible for studying the market needs, and follow-up on advertising, media and assess the effectiveness of the activities, and monitor the level of sales and organization.
- **Maintenance employees:** will keep the machines working and provide periodic maintenance and repair when broken.
- **Mechanical Technician:** Find out mechanical breakdowns of industrial machinery and equipment, and supplies work materials according to quantities and specifications.
- **Electrical technician:** Applies his skills and information in the field of maintenance for electrical appliances, motors, transformers, cutouts and switches.
- **Quality auditors:** Check the effectiveness of administrative performance and work procedures in accordance with the legislations in force.
- **Warehouse staff:** responsible for arranging and storing the final product in warehouses and maintaining them.

Technical Requirements Analysis

Land and Costruction works:

The project will be constructed on a plot area of 8000 m². The following table shows the plot area needed:

Table 11 : Required Land

Land	Area(m ²)
The Required plot area	8000

The construction works required for the project are represented as follows:

Table12 : Required construction works

Required construction work	Area (m2)
Hangar works	2,000
Construction works	500
land	300

The following table shows the machinery and equipment needed for the project as follows:

Table 13 :Machinery and equipment

Machinery and equipment	Number
Raw Materials Slio	1
Furnace	1
Fire polishing machine	1
Relay Machine	1
Cutting Line	1
Storage Equipment	3

Furniture

The project needs to the following furniture:

Table 14: Furniture and furnishing

furniture and furnishings	Number	Total cost / JD
furniture and furnishings	–	18,000

Vehicles and Transportation

The project needs to the following transport methods:

Table 15: Required vehicles and transportation

Vehicles and transportation	Number
Transportation vehicles	3

Required Raw Materials

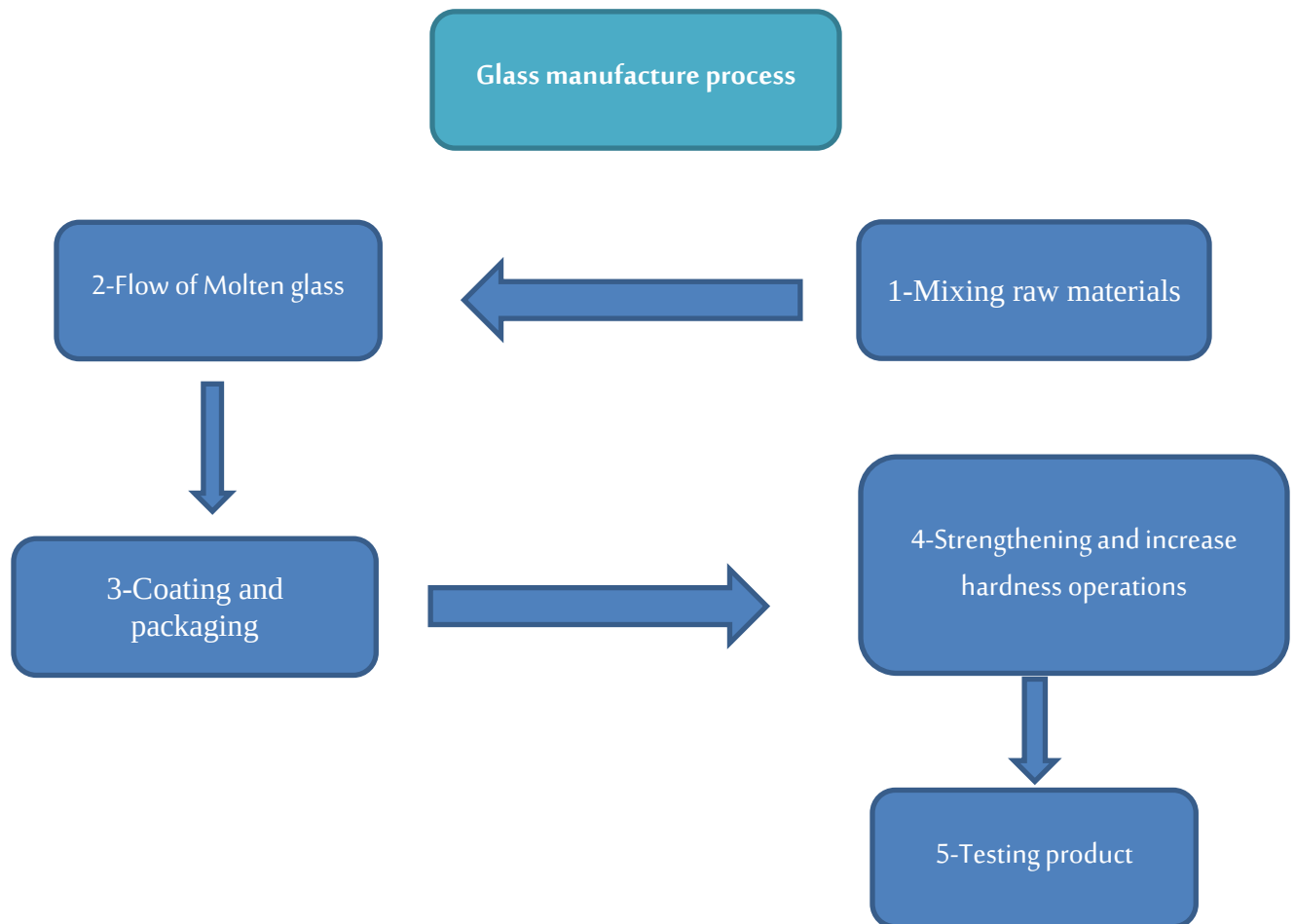
Glass needs a range of material that enters its composition. These materials are:

Table16 :Annual raw material qantity required

Materials	Required Quantity (Ton/Yea)	Price (JOD/ton)
Silica Sand	6,000	65
Soda	1,000	110
Others	1,000	25

Production Process Description

The glass manufacturing processes are illustrated as follows:



Manufacturing process details:

1. Mix the raw materials (silica, sand, calcium, soda, oxides and magnesium) with each other in weighted proportions and then melt them at a temperature of 1500 °C.
2. The molten glass flow from the furnace towards a bath of molten tin. At this stage the required thickness of the glass panels is determined.

3. Coating and packaging, at this stage, the metal oxides are directly applied on the glass, while the glass is still hot, in (tempering lehr). It is worth mentioning that these materials can be added to the mix to create the different colors of glass.
4. Operations of strengthening and increasing the rigidity of the glass plates.
5. Final testing for product quality.

5. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 12.81% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below

Table 17:General Information

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 18:Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 19: Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 20: Assumptions of expenses

Assumptions of expenses	
Expenses of raw materials of income (food and drink)	25.00%
Service benefit expenses from revenues	5.00%
Maintenance expenses from revenues	1.00%
Consumer expenses from revenues	4.00%
Insurance expenses from the value of assets	0.50%
Marketing expenses of revenue	0.50%

Table 21: Income Tax Assumptions

Income Tax Assumptions	Value
(1) Average Income Tax in Jordan	5%
(2) Income Tax Deduction	0%
Income Tax after Deduction	5%
(3) Compulsary Reserve Percentage	10%
Other Assumptions	القيمة
(1) Annual Monthly Salaries have been calculated after multiplying monthly salaries by:	16

Table 22: Risk Premium

Risk Premium	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	9.48%
(3) Market Risk Premium	10.12%
Income Tax Rate	5.00%
(4) Beta	1
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 23: Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premium	%10.12
Income Tax Rate	%5
Bets	1.00
Equity	1,085,483

Loans Value	1,085,483
Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%9.01
Loans Value	1,085,483
Loans Percentage	%50
Equity	
Cost of Equity	%16.6
Equity Value	1,085,483
Equity Percentage	%50
Gross	
Project Value	2,170,965
Project Percentage	%100
Weighted Average Cost of Capital	%12.81

Capital Expenditures

The estimated cost of the project is 2,170,965JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses, working capital and vehicle requirement. The following tables summarize the details of these expenses:

Table 24:Land Capital Expenditure

Land	Jordan Dinar/Meters) Cost	Area (M2)	الإجمالية / دينار Cost
Required Land	10	8,000	80,000
Cost			80,000

Table 25:Construction Work Capital Expenditure

Description	Cost (JOD per Squared	Area	Gross Cost/ JD
Hanger	100	2,000	200,000
Building Construction	250	500	125,000
Paving	12	300	3,600
Other Works			25,000
Total			353,600

Table 26: Machines and equipment Capital Expenditure

Description	Number	Unit Cost / JD	Gross Cost/ JD
Raw Materials Slio	1	235,000	235,000
Furnace	1	400,000	400,000
Fire polishing machine	1	200,000	200,000
Relay Machine	1	150,000	150,000
Cutting Line	1	175,000	175,000
Storage Equipment	3	40,000	120,000
Total (JD)			1,280,000

Table 27:Furniture Capital Expenditure

Description	Number	Unit Cost / JD	Gross Cost/ JD
Furniture	-	-	18,000
Total			18,000

Table 28:Transporation and Vehicles Capex

Transporation and Vehicles	No	Unit Cost/JOD	Total Cost/JOD
Pick up	3	25,000	75,000
Total			75,000

Pre Operating Expenses

Consist of a set of expenses that are expended before the start of the project and before any revenue is realized. It consist of Government fees, licenses, marketing expenses, training and transportation expenses which are estimated as follows:

Table 29:Pre-operating expenses

Pre-operating expenses	Estimated cost (JD)
Governmental fees	1,500
Transportation	1,500
Marketing	1,000
Training	1,000
Total	5,000

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 30: Working Capital

Working Capital	No. of Months	%	Estimated Cost (JD)
Operating expenses	3	%25	113,493
General and administrative expenses	3	%25	4,410
Marketing expenses	3	%25	1,103
Indirect Salaries	3	%25	43,000
Total			162,005

Table 31: Expenses Summary

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Land	80,000	10%	88,000	%4.1
Construction works	353,600	10%	388,960	%17.9
Machinery & Equipment	1,280,000	10%	1,408,000	%64.9
Furniture	18,000	10%	19,800	%0.9
Vehicles	75,000	10%	82,500	%3.8
Pre-Operating Expenses	5,000	10%	5,500	%0.3
Working capital	162,005	10%	178,205	%8.2
Total (JD)	1,973,605		2,170,965	100%

Source of funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (2,170,965) JD, where 50% of the project will be financed through loans (1,085,483) JD, and the remaining (50%) through (Equity) with (1,085,483) JD. The following table summarizes the general structure of the required financing:

Table 32: Sources of Funding

Funding Sources	Amount	Percentage
Equity	1,085,483	%50
Loans	1,085,483	%50
Total	2,170,965	%100
Use of Fund:	Value	Ratio
Capital Expenditures	1,987,260	%91.54
Pre-Operating Expenses	5,500	%0.25
Working Capital	178,205	%8.21
Total	2,170,965	%100.00

Operating Expenses

The following table summarizes the expected operating expenses for the proposed project (2018-2027):

Table 33: Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials and Packaging	220,500	275,625	312,375	312,375	312,375	349,125	349,125	367,500	367,500	367,500
Utilities from Revenues	44,100	55,125	62,475	62,475	62,475	69,825	69,825	73,500	73,500	73,500
Maintenance from Revenues	8,820	11,025	12,495	12,495	12,495	13,965	13,965	14,700	14,700	14,700
Disposables from Revenues	35,280	44,100	49,980	49,980	49,980	55,860	55,860	58,800	58,800	58,800
Direct Salaries	104,000	108,264	112,703	117,324	122,134	127,141	132,354	137,781	143,430	149,310
Others	41,270	49,414	55,003	55,465	55,946	61,592	62,113	65,228	65,793	66,381
Total	453,970	543,553	605,031	610,114	615,405	677,508	683,242	717,509	723,723	730,191

General and Administrative Expenses

The following table summarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table34 :General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Hospitality	1,000	1,030	1,061	1,093	1,126	1,159	1,194	1,230	1,267	1,305
Stationary and Printing	1,500	1,545	1,591	1,639	1,688	1,739	1,791	1,845	1,900	1,957
Transportation	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Training	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Insurance	9,936	10,234	10,541	10,858	11,183	11,519	11,864	12,220	12,587	12,965
Miscellaneous	1,604	1,652	1,701	1,752	1,805	1,859	1,915	1,972	2,031	2,092
Total	17,640	18,169	18,714	19,276	19,854	20,450	21,063	21,695	22,346	23,016

Marketing Expenses

Table 35:Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	4,410	5,513	6,248	6,248	6,248	6,983	6,983	7,350	7,350	7,350

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 36: Expected monthly salaries and wages for the years (2018-2027)

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect Employees										
General Manager	3,000	3,123	3,251	3,384	3,523	3,668	3,818	3,974	4,137	4,307
Engineers	750	781	813	846	881	917	954	994	1,034	1,077
Human Resources Manager	1,000	1,041	1,084	1,128	1,174	1,223	1,273	1,325	1,379	1,436
Labortary worker	400	416	433	451	470	489	509	530	552	574
Sales representatives	400	416	433	451	470	489	509	530	552	574
Financial Manager	1,000	1,041	1,084	1,128	1,174	1,223	1,273	1,325	1,379	1,436
Accountant	500	521	542	564	587	611	636	662	690	718
Human Resources Employees	450	521	542	564	587	611	636	662	690	718
Direct Employees										
Quality Control Managers	700	729	759	790	822	856	891	927	965	1,005
Maintenance technician	400	416	433	451	470	489	509	530	552	574
Warehouse manager	700	729	759	790	822	856	891	927	965	1,005

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Electrical technician	500	521	542	564	587	611	636	662	690	718
Mechanical technician	500	521	542	564	587	611	636	662	690	718
Quality Control Engineers	600	625	650	677	705	734	764	795	827	861

Table 37: Expected Annual Salaries

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect Employees										
General Manager	48,000	49,968	52,017	54,149	56,369	58,681	61,087	63,591	66,198	68,912
Engineers	12,000	12,492	13,004	13,537	14,092	14,670	15,272	15,898	16,550	17,228
Human Resources Manager	16,000	16,656	17,339	18,050	18,790	19,560	20,362	21,197	22,066	22,971
Labortary worker	32,000	33,312	34,678	36,100	37,580	39,120	40,724	42,394	44,132	45,942
Sales representatives	32,000	33,312	34,678	36,100	37,580	39,120	40,724	42,394	44,132	45,942
Financial Manager	16,000	16,656	17,339	18,050	18,790	19,560	20,362	21,197	22,066	22,971
Accountant	16,000	16,656	17,339	18,050	18,790	19,560	20,362	21,197	22,066	22,971
Human Resources Employees	7,200	16,656	17,339	18,050	18,790	19,560	20,362	21,197	22,066	22,971
Total indirect salaries	172,000	179,052	186,393	194,035	201,991	210,272	218,893	227,868	237,211	246,936
Direct Employees										
Quality Control Managers	22,400	23,318	24,274	25,270	26,306	27,384	28,507	29,676	30,893	32,159

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Maintenance technician	19,200	19,987	20,807	21,660	22,548	23,472	24,435	25,436	26,479	27,565
Warehouse manager	11,200	11,659	12,137	12,635	13,153	13,692	14,254	14,838	15,446	16,080
Electrical technician	16,000	16,656	17,339	18,050	18,790	19,560	20,362	21,197	22,066	22,971
Mechanical technician	16,000	16,656	17,339	18,050	18,790	19,560	20,362	21,197	22,066	22,971
Quality Control Engineers	19,200	19,987	20,807	21,660	22,548	23,472	24,435	25,436	26,479	27,565
Total direct salaries	104,000	108,264	112,703	117,324	122,134	127,141	132,354	137,781	143,430	149,310
Total	276,000	287,316	299,096	311,359	324,125	337,414	351,248	365,649	380,640	396,247

Depreciations

The following tables illustrate cost of Capex and Depreciation rates:

Table 38: Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	88,000	%0.0	%0.0
Construction Works	388,960	%5.0	%0.0
Machinaries	1,408,000	%10.0	%2.0
Furniture	19,800	%10.0	%5.0
Vehicles	82,500	%10.0	%0.0
Total	1,987,260		

Table 39: Depreciations and Additions on Construction Works

Capex, Annual Additions and Depreciations	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	1,987,260	2,016,410	2,046,173	2,076,562	2,107,592	2,139,276	2,171,631	2,204,670	2,238,410	2,272,867
Total Depreciation	170,478	173,393	176,369	179,408	182,511	185,680	188,915	192,219	195,593	199,039
Total Accumulated	170,478	343,871	520,240	699,648	882,160	1,067,839	1,256,754	1,448,973	1,644,566	1,843,605
Total Additions	0	29,150	29,763	30,389	31,030	31,685	32,354	33,039	33,740	34,457
Total Net Book Values	1,816,782	1,672,539	1,525,932	1,376,913	1,225,432	1,071,437	914,876	755,697	593,844	429,262

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project .

Table 40:Loan Details

Loan Value	1,085,483
Annual Interest Rate	9.48%
Loans Period	5
Grace Period	0
Loan Starts at year:	2018
Annual Payment	282,554
Number of Payments	5

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2018				1,085,483
2019	282,554	102,904	179,650	905,833
2020	282,554	85,873	196,681	709,152
2021	282,554	67,228	215,326	493,826
2022	282,554	46,815	235,739	258,087
2023	282,554	24,467	258,087	0

Income Statement

Table 41: Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	882,000	1,102,500	1,249,500	1,249,500	1,249,500	1,396,500	1,396,500	1,470,000	1,470,000	1,470,000
Gross Operating	882,000	1,102,500	1,249,500	1,249,500	1,249,500	1,396,500	1,396,500	1,470,000	1,470,000	1,470,000
Operating Expenses	-453,970	-543,553	-605,031	-610,114	-615,405	-677,508	-683,242	-717,509	-723,723	-730,191
Gross Operating Profit	428,030	558,947	644,469	639,386	634,095	718,992	713,258	752,491	746,277	739,809
<i>Gross Profit Percentage</i>	<i>%49</i>	<i>%51</i>	<i>%52</i>	<i>%51</i>	<i>%51</i>	<i>%51</i>	<i>%51</i>	<i>%51</i>	<i>%51</i>	<i>%50</i>
Salaries and Benefits	-172,000	-179,052	-186,393	-194,035	-201,991	-210,272	-218,893	-227,868	-237,211	-246,936
General and	-17,640	-18,169	-18,714	-19,276	-19,854	-20,450	-21,063	-21,695	-22,346	-23,016
Marketing Expenses	-4,410	-5,513	-6,248	-6,248	-6,248	-6,983	-6,983	-7,350	-7,350	-7,350
Pre Operating Expenses	-5,500									
Gross Indirect Expenses	-199,550	-202,734	-211,355	-219,558	-228,092	-237,704	-246,939	-256,913	-266,906	-277,302
Income before Interest,	228,480	356,213	433,115	419,828	406,003	481,288	466,319	495,578	479,371	462,506
Fixed Assets	-170,478	-173,393	-176,369	-179,408	-182,511	-185,680	-188,915	-192,219	-195,593	-199,039
Income before Tax and	58,002	182,820	256,745	240,420	223,492	295,608	277,404	303,359	283,778	263,468
Bank Interests	-102,904	-85,873	-67,228	-46,815	-24,467	-0	-0	-0	-0	-0
Income Before Tax	-44,902	96,948	189,518	193,605	199,025	295,608	277,404	303,359	283,778	263,468
Income Tax	0	-2,602	-9,476	-9,680	-9,951	-14,780	-13,870	-15,168	-14,189	-13,173

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Profit	-44,902	94,345	180,042	183,925	189,074	280,828	263,534	288,191	269,589	250,294
<i>Net Profit Percentage</i>	<i>%5-</i>	<i>%9</i>	<i>%14</i>	<i>%15</i>	<i>%15</i>	<i>%20</i>	<i>%19</i>	<i>%20</i>	<i>%18</i>	<i>%17</i>
Compulsory Reserves	0	-9,435	-18,004	-18,392	-18,907	-28,083	-26,353	-28,819	-26,959	-25,029
Retained Earnings	-44,902	40,009	202,047	367,579	537,746	790,491	1,027,671	1,287,043	1,529,673	1,754,938

Expected Cashflows Statement

The table below summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 42: Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating Activities										
Net Profit	-44,902	94,345	180,042	183,925	189,074	280,82	263,534	288,191	269,589	250,294
Bank Interests	102,904	85,873	67,228	46,815	24,467	0	0	0	0	0
Depreciation	170,478	173,393	176,369	179,408	182,511	185,68	188,915	192,219	195,593	199,039
Total Operating Cashflows before Additions to Working Capital	228,480	353,611	423,639	410,148	396,052	466,507	452,449	480,410	465,182	449,333
Inventory (Increase/Decrease)	-18,375	-4,594	-3,063	0	0	-3,063	0	-1,531	0	0

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Accounts Receivables	0	0	0	0	0	0	0	0	0	0
Accounts Payables (Increase/Decrease)	39,690	5,858	4,204	864	900	4,310	975	2,702	1,056	1,100
Working Capital (Increase/Decrease)	21,315	1,264	1,141	864	900	1,248	975	1,170	1,056	1,100
Net Cashflows from Operating Activities	249,795	354,875	424,780	411,012	396,951	467,755	453,424	481,581	466,238	450,432
Cashflows from Investments Activities										
Fixed Assets (Procurement)	1,987,260	-29,150	-29,763	-30,389	-31,030	-31,685	-32,354	-33,039	-33,740	-34,457
Net Cashflows from Investments Activities	1,987,260	-29,150	-29,763	-30,389	-31,030	-31,685	-32,354	-33,039	-33,740	-34,457
Cashflows from Financing Activities										
Capital	1,085,483									
Loan Amortization	-179,650	196,681	215,326	235,739	258,087	0	0	0	0	0
Bank Interest Rate	-102,904	-85,873	-67,228	-46,815	-24,467	-0	-0	-0	-0	-0
Loans	1,085,483	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing Activities	1,888,412	282,554	282,554	282,554	282,554	0	0	0	0	0
Net (Increase/Decrease) in Cash	150,947	43,172	112,464	98,069	83,368	436,07	421,069	448,541	432,498	415,976
Cashflows at the Beginning of Period	0	150,947	194,118	306,582	404,651	488,01	924,089	1,345,15	1,793,69	2,226,19
Cashflows at the End of Period	150,947	194,118	306,582	404,651	488,019	924,08	1,345,15	1,793,69	2,226,19	2,642,17

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 43: Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current Assets										
Cash	150,947	194,118	306,582	404,651	488,019	924,089	1,345,158	1,793,699	2,226,198	2,642,174
Inventory	18,375	22,969	26,031	26,031	26,031	29,094	29,094	30,625	30,625	30,625
Accounts Receivable	0	0	0	0	0	0	0	0	0	0
Total current Assets	169,322	217,087	332,613	430,682	514,050	953,183	1,374,252	1,824,324	2,256,823	2,672,799
Non Current Assets										
Fixed Assets (net)	1,816,782	1,672,539	1,525,932	1,376,913	1,225,432	1,071,437	914,876	755,697	593,844	429,262
Total Non Current Assets	1,816,782	1,672,539	1,525,932	1,376,913	1,225,432	1,071,437	914,876	755,697	593,844	429,262
Total Assets	1,986,104	1,889,626	1,858,546	1,807,596	1,739,482	2,024,620	2,289,128	2,580,021	2,850,666	3,102,060
Liabilities										
Current Liabilities										
Remaining amount of Loan	39,690	45,548	49,751	50,616	51,515	55,825	56,800	59,501	60,558	61,658

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total current Liabilities	196,681	215,326	235,739	258,087	0-	0-	0-	0-	0-	0
Non Current Liabilities	236,371	260,874	285,490	308,703	51,515	55,825	56,800	59,501	60,558	61,658
Long Terms Loans										
Total Long Term Liabilities	709,152	493,826	258,087	0	0	0	0	0	0	0
Total Liabilities	709,152	493,826	258,087	0	0	0	0	0	0	0
Owners Equity	945,523	754,700	543,577	308,703	51,515	55,825	56,800	59,501	60,558	61,658
Shareholders Contributions	1,085,483	1,085,483	1,085,483	1,085,483	1,085,483	1,085,483	1,085,483	1,085,483	1,085,483	1,085,483
Statutory Reseve	0	9,435	27,439	45,831	64,739	92,821	119,175	147,994	174,953	199,982
Retained Profits	44,902-	40,009	202,047	367,579	537,746	790,491	1,027,671	1,287,043	1,529,673	1,754,938
Total Equity	1,040,581	1,134,926	1,314,968	1,498,893	1,687,967	1,968,795	2,232,328	2,520,520	2,790,109	3,040,403
Total Liabilities and Equity	1,986,104	1,889,626	1,858,546	1,807,596	1,739,482	2,024,620	2,289,128	2,580,021	2,850,666	3,102,060

Feasibility Indicators

There is more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 44: Free Net Cash flows Table

Net Cashflows	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final Value
Net Free	(2,170,965)	255,295	325,725	395,017	380,623	365,922	436,070	421,069	448,541	432,498	415,976	4,908,827
Discount	1.00	0.70	0.62	0.55	0.49	0.43	0.38	0.34	0.30	0.27	0.24	0.24
Net Present	(2,170,965)	177,814	201,101	216,182	184,646	157,353	166,220	142,273	134,342	114,824	97,895	1,155,229

Table below illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 45: Payback Period

Payback	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free	-2,170,965	255,295	325,725	395,017	380,623	365,922	436,070	421,069	448,541	432,498	415,976
Project Value	2,170,965	1,915,671	1,589,945	1,194,928	814,305	448,384	12,313	-408,756	-857,297	-1,289,795	-1,705,771
Payback	6	1	1	1	1	1	1	0	0	0	0

Table 46: Financial Analysis Results

Description	
Weighted Average Cost of Capital (WACC)	%12.81
Net Present Value for Cashflows	576,912
Payback Period	6
Internal Rate of Return	%20.06

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes the declined in revenues, changes in total investment costs, and increase in operating costs with 10%, which indicate that the project is feasible.

The following table shows that the project will be feasible if the revenues declined with 10% or the operating costs increase by 10%.

Table 47: Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%20.06	6	12.81%
Revenues declined by 10%	%16.22	7	12.81%
Operating Expenses Increased by 10%	%19.93	6	12.81%

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 576,912 Jordan Dinars considering that the project provides 30 Job Opportunities for the governorate residents.